

1. Recursive Array Sum: Write a function that accepts an array of integers and a number indicating the number of elements as arguments. The function should recursively calculate the sum of all the numbers in the array. Demonstrate the function in a driver program.
2. Recursive Multiplication: Write a recursive function that accepts two arguments into the parameters x and y . The function should return the value of x times y . Remember, multiplication can be performed as repeated addition: $7 * 4 = 4 + 4 + 4 + 4 + 4 + 4 + 4$
3. Recursive Power Function: Write a function that uses recursion to raise a number to a power. The function should accept two arguments: the number to be raised and the exponent. Assume that the exponent is a nonnegative integer. Demonstrate the function in a program.
4. Sum of Numbers: Write a function that accepts an integer argument and returns the sum of all the integers from 1 up to the number passed as an argument. For example, if 50 is passed as an argument, the function will return the sum of 1, 2, 3, 4, ... 50. Use recursion to calculate the sum. Demonstrate the function in a program.
5. isMember Function Write a recursive Boolean function named isMember. The function should accept two arguments: an array and a value. The function should return true if the value is found in the array, or false if the value is not found in the array. Demonstrate the function in a driver program.
6. String Reverser: Write a recursive function that accepts a string object as its argument and prints the string in reverse order. Demonstrate the function in a driver program.
7. Palindrome Detector: A palindrome is any word, phrase, or sentence that reads the same forward and backward. Here are some well-known palindromes:
Able was I, ere I saw Elba
A man, a plan, a canal, Panama
Desserts I stressed
Kayak
Write a bool function that uses recursion to determine if a string argument is a palindrome. The function should return true if the argument reads the same forward and backward. Demonstrate the function in a program.
8. Ackermann's Function: Ackermann's Function is a recursive mathematical algorithm that can be used to test how well a computer performs recursion. Write a function $A(m, n)$ that solves Ackermann's Function. Use the following logic in your function:
If $m = 0$ then return $n + 1$
If $n = 0$ then return $A(m-1, 1)$
Otherwise, return $A(m-1, A(m, n-1))$

Test your function in a driver program that displays the following values:
A(0, 0) A(0, 1) A(1, 1) A(1, 2) A(1, 3) A(2, 2) A(3, 2)

9. Write an iterative version (using a loop instead of recursion) of the factorial function.
10. Suppose that `intArray` is an array of integers, and `length` specifies the number of elements in `intArray`. Also, suppose that `low` and `high` are two integers such that $0 \leq \text{low} < \text{length}$, $0 \leq \text{high} < \text{length}$, and $\text{low} < \text{high}$. That is, `low` and `high` are two indices in `intArray`. Write a recursive definition that reverses the elements in `intArray` between `low` and `high`.
11. Describe a way to use recursion to compute the sum of all the elements in a $n \times n$ (two-dimensional) array of integers.
12. Describe a recursive algorithm for finding the maximum element in an array `A` of n elements.